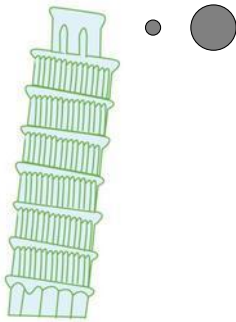


Newton's Law Concept Questions

Constancy of Free Fall

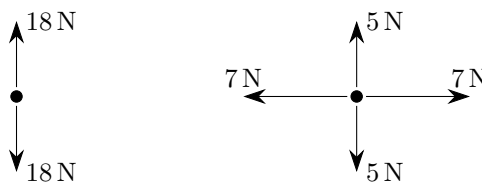
In the famous Leaning Tower of Pisa experiment, Galileo dropped two balls from the top of the tower. Let's say that one was 5 kg and the other was 500 kg

- (a) Calculate the weight of the 5-kg ball.
- (b) Calculate the weight of the 500-kg ball.
- (c) Newton's Second Law says that the ball with more force (*i.e.* more weight) should have a greater acceleration. How can both balls have the same acceleration?

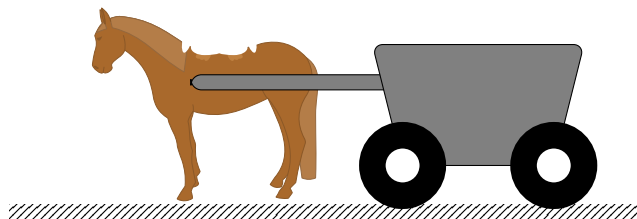


Horse and Cart

Calculate the net force in each of the free-body diagrams that have equal and opposite forces:

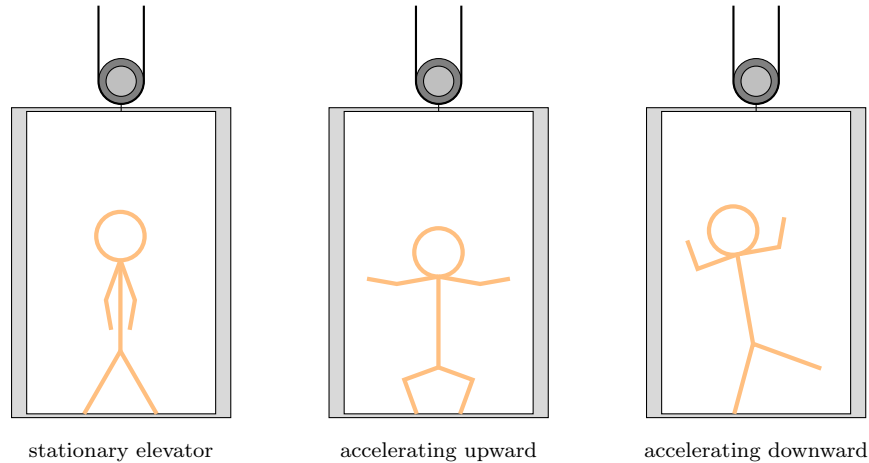


A horse pulls a cart with a force of 400 Newtons forward. However, according to Newton's Third Law, whenever the horse pulls the cart, there is an equal and opposite reaction of the cart pulling the horse of 400 N backwards. Why do these two forces not cancel each other out giving to a net force of zero?



Elevator Effect

When do you are riding in an elevator, there are some times that you feel heavier and some times that you feel lighter.



Which force corresponds to “how heavy” you feel?

How could you use an elevator to simulate weightlessness?

Example Problem Let's say that a 75-kg person is standing in an elevator that is acceleration downward at 2.1 m/s^2 . What is the normal force acting on this person? Do they feel heavier or lighter than usual?